

DATA AND AI IN ECONOMICS · TU DORTMUND

Clearing the Air

Causal Policy Impacts, Meteorological Regimes & Machine-Learning Predictors of PM2.5 Across Five Chinese Macroeconomic Hubs

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Group G

Session: Energy, Climate & Infrastructure

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Motivation & Economic Relevance

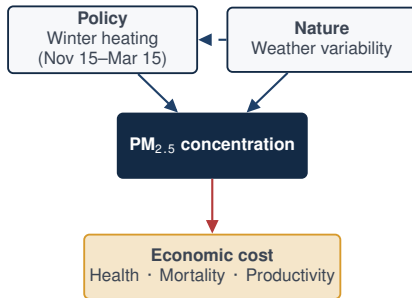
Why disentangling policy from weather matters

Research question

To what extent is the corrected northern winter-heating regime in Beijing and Shenyang associated with higher hourly $PM_{2.5}$, relative to non-heating-zone cities (Shanghai, Guangzhou, Chengdu), once meteorological confounders are controlled for — and how well can non-linear ML models forecast these dynamics?

Why it matters

- Air pollution is a severe **negative externality** of industrialization: excess mortality, health costs, and lower urban worker productivity
- Policy-driven heating emissions and natural weather variation are entangled — separating them is essential for designing **efficient, targeted** clean-air regulation
- Findings inform the **timing** of green-energy transitions without derailing regional growth



Weather drives both heating demand and pollution dispersion — the analysis estimates the adjusted heating-regime association.

Data & Methodology

Hourly panel, 2010–2015, three complementary method blocks

Data

- **Panel:** 175,993 hourly city-observations, Jan 2010–Dec 2015, 5 cities (Beijing, Shenyang [*heating zone*]; Shanghai, Guangzhou, Chengdu [*non-heating zone*])
- **Sources:** UCI 5-Cities PM_{2.5} dataset (Liang et al., 2016); Open-Meteo reanalysis weather; EDGAR city-year emissions (background only)
- **Target:** PM_{2.5} ($\mu\text{g}/\text{m}^3$); **Treatment:** heating_regime (binary, North & Nov 15–Mar 15)
- **Cleaning:** limited short-gap PM_{2.5} forward fill, then dropping remaining missing target rows; Open-Meteo used for consistent weather controls

Three method blocks

1. Causal Inference — DAG/backdoor adjustment on weather & city/year/month/hour fixed effects; clustered SE; validated with placebo, random-common-cause & subset refuter-style checks

2. Supervised Learning — Random Forest delta vs. OLS/Ridge & lag baselines; **chronological** split with 2014 validation and untouched 2015 holdout; RMSE / MAE / R^2

3. Unsupervised Learning — K-Means ($k=4$ retained for interpretability after k -checks) on standardized, PCA-reduced weather features; regimes mapped back onto PM_{2.5} for validation

Results I — Heating-Regime Association

Backdoor-adjusted regression, robustness-checked

Backdoor-adjusted heating-regime association

+21.33

$\mu\text{g}/\text{m}^3 \text{ PM}_{2.5}$

Clustered SE = 5.99 · 95% CI [9.59, 33.07] · $p = 0.0004$

After controlling for weather and city / year / month / hour fixed effects, the corrected northern heating-regime period is associated with substantially higher $\text{PM}_{2.5}$.

Robustness checks (refuter-style checks)

- **Placebo treatment:** mean effect ≈ -0.06 (near zero) → no strong placebo signal
- **Random common cause:** coefficient shifts by only 0.01% → insensitive to added noise
- **25× data-subset resampling:** estimates range 16.0–27.9, mean 22.3 → stable across repeated subsamples
- **EDGAR city-year background controls:** 21.33 → 20.75 (−0.58) → robust to background emissions

Full robustness table in Backup B.

Results II — Forecasting & Weather Regimes

Predictive accuracy and latent atmospheric clusters

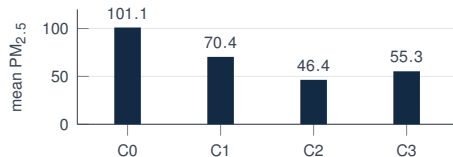
Supervised learning (2015 holdout)

- Best model: **RF (Δ -PM)** — RMSE 11.65, MAE 6.14, $R^2 = 0.966$
- +10.6% RMSE reduction vs. lag-1 persistence;
+4.0% vs. OLS baseline — **short of the proposal's 15% target** (bootstrap: $P(\geq 15\%) \approx 0.3\%$)
- All 5 cities show a 2015 KS-distribution shift → holdout is a *hard*, out-of-distribution test, not a clean i.i.d. split

Full model comparison in Backup C.

Unsupervised learning ($k=4$ weather regimes)

- PCA: PC1+PC2 explain **50.8%** of weather variance
- Regime 0 (**cold, low boundary layer**) → highest burden: mean $PM_{2.5} = 101.1$
- Regime 2 (**warm, humid/rainy conditions**) → lowest burden: mean $PM_{2.5} = 46.4$
- Kruskal–Wallis: clusters built on weather *alone* still separate $PM_{2.5}$ significantly ($p < 0.001$)



Limitations & Synthesis

What the three lenses agree on — and where caution is needed

Limitations

- Analysis is at **city-hour** level — station-level heterogeneity within cities is masked
- EDGAR emissions are **annual, city-level** background figures, not true hourly emissions (sensitivity check only)
- Causal estimate is identified **under the DAG / backdoor assumptions** — no exogenous policy shock is exploited
- 2015 ML holdout shows a significant **distribution shift** in all 5 cities (harder than an i.i.d. test)
- The proposal's **15% RMSE** improvement bar for ML is not met — gains are real but modest

Synthesis

- The causal lens shows that the corrected heating-regime period is associated with about $+21 \mu\text{g}/\text{m}^3$ higher $\text{PM}_{2.5}$ after weather and fixed-effect adjustment
- **Cold, stagnant** regimes show the highest $\text{PM}_{2.5}$ burden — policy timing and meteorology should be interpreted together
- ML forecasts add **modest** value beyond simple persistence — dynamics are strongly shaped by autocorrelation and regime-level conditions
- **Policy implication:** clean-air regulation should be **regime-aware** (timing $\times$ weather), not purely calendar-based

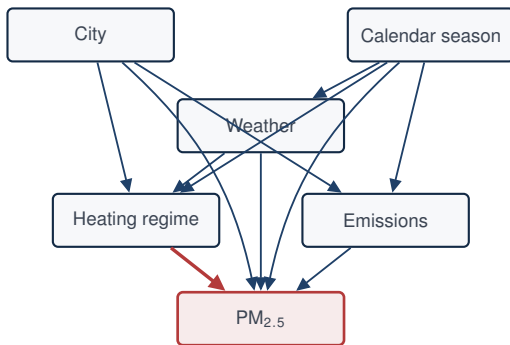
Literature & Data Sources

- Liang, X., Li, S., Zhang, S., Huang, H., & Chen, S. X. (2016). PM_{2.5} data reliability, consistency, and air quality assessment in five Chinese cities. *Journal of Geophysical Research: Atmospheres*, 121(17), 10,220–10,236.
- UCI Machine Learning Repository — *PM_{2.5} Data of Five Chinese Cities* dataset.
- Open-Meteo Historical Weather API — reanalysis meteorological data.
- EDGAR — Emissions Database for Global Atmospheric Research, European Commission Joint Research Centre.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press.
- Pedregosa, F., et al. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Backup Slides

Backup A — Causal Identification Strategy

Directed acyclic graph and backdoor adjustment set



Backdoor controls

- TEMP, PRES, HUMI, DEWP, Iws
- precipitation_log1p, wind_sin, wind_cos
- boundary_layer_height, cloud_cover
- City, year, month & hour fixed effects

Treatment / Outcome

- Treatment: heating_regime
- Outcome: PM (PM_{2.5})
- Lagged PM excluded from the causal model; used only in supervised forecasting
- EDGAR emissions used as background sensitivity, not as the main causal control

Backup B — Causal Robustness Checks

Check	Result
Main backdoor model	$\beta = 21.329$, clustered SE = 5.990 (city $\times$ month, 250 clusters), 95% CI [9.590, 33.068], $p = 0.0004$
Placebo treatment (25 draws)	mean = -0.057 , sd = 0.360, range [-0.638 , 0.755]
Random common cause	$\beta = 21.332$ ($\Delta = 0.0026$, 0.01% relative change)
Data-subset refuter (25 $\times$, 80% blocks)	mean = 22.283, sd = 3.031, range [16.043, 27.850]
EDGAR city-year sensitivity	$\beta = 20.752$ ($\Delta = -0.577$ vs. main model)

Rows used: 175,993 · Clusters: city $\times$ month (250) · Treatment: `heating_regime` · Outcome: PM (PM_{2.5})

Backup C — Supervised Model Comparison (2015 Holdout)

Model	RMSE	MAE	R^2	% red. vs. Lag-1	% red. vs. OLS
RF delta	11.649	6.138	0.966	10.6	4.0
HGB delta	11.663	6.185	0.966	10.5	3.9
Ridge engineered	12.134	6.664	0.963	6.9	0.0
OLS delta	12.137	6.669	0.963	6.8	0.0
OLS engineered	12.137	6.669	0.963	6.8	0.0
RF engineered	12.680	6.363	0.960	2.7	-4.5
Lag-1 persistence	13.027	6.884	0.958	0.0	-7.3
HGB engineered	14.704	6.463	0.946	-12.9	-21.1
Lag-24 persistence	63.522	35.876	-0.006	-387.6	-423.4

Final train: 2010–2014 (127,670 rows) · Test: 2015 (43,677 rows, untouched) · 73 model features
Bootstrap (1,000 city×month blocks): $P(\text{RMSE reduction of RF delta vs. Lag-1} \geq 15\%) = 0.003$

Backup D — Weather-Regime Profiles & City Composition

Cluster physical & pollution profile

	C0	C1	C2	C3
TEMP (°C)	2.91	22.72	19.48	18.61
HUMI (%)	58.92	75.95	83.82	38.45
lws (m/s)	8.00	7.24	13.34	14.92
Boundary layer (m)	270.7	337.5	506.7	1415.1
Cloud cover (%)	35.81	60.76	94.94	40.81
Mean PM _{2.5}	101.07	70.39	46.40	55.33
n (hours)	53,075	69,236	28,239	25,443

City × regime distribution (row %)

City	C0	C1	C2	C3
Beijing	44.6	31.7	3.8	19.9
Shenyang	40.9	27.5	7.5	24.1
Shanghai	23.0	32.9	31.3	12.8
Chengdu	26.9	52.6	13.6	6.9
Guangzhou	10.2	54.8	27.6	7.4

C0 = cold / low boundary layer (stagnant, highest PM_{2.5}); C2 = warm / humid-rainy conditions (lowest PM_{2.5}). PCA: PC1+PC2 explain 50.8% of weather variance; $k = 4$ retained for interpretability after k -checks.